



MALAPPURAM SAHODAYA APTITUDE TEST
EXAMINATION 2023-24
DISTRICT LEVEL

Class: 10

Date: 4/11/23

Marks: 100

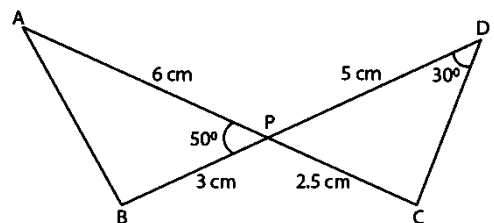
Name of the Student:

Time: 2 Hrs

General Instructions:

1. Candidate will be supplied with separate Question Paper and Answer Sheet.
2. Answer the questions in the OMR sheet by shading the appropriate bubble.
3. Shade the bubbles with black ball pen/ HB pencil.
4. Question number 1 to 30 carries two marks and 31 to 70 one mark each.
5. For each question below, four options are given. One of them is correct answer. Make your choice and shade the correct box.

1. $n^2 - 1$ is divisible by 8, if n is
 - a. A natural number
 - b. An integer
 - c. An odd integer
 - d. An even integer
2. The 11th term of the AP: $-5, \left(-\frac{5}{2}\right), 0, \frac{5}{2}, \dots$ is
 - a. -20
 - b. 20
 - c. -30
 - d. 30
3. A quadratic polynomial, whose zeroes are -3 and 4, is
 - a. $x^2 - x + 12$
 - b. $x^2 + x + 12$
 - c. $\left(\frac{x^2}{2}\right) - \left(\frac{x}{2}\right) - 6$
 - d. $2x^2 + 2x - 24$
4. Which of the following cannot be the probability of an event?
 - a. $\frac{1}{3}$
 - b. 0.1
 - c. 3%
 - d. $\frac{17}{16}$
5. AOBC is a rectangle whose three vertices are A(0, 3), O(0, 0) and B(5, 0). The length of its diagonal is
 - a. 5
 - b. 3
 - c. $\sqrt{34}$
 - d. 4
6. If the lengths of the diagonals of rhombus are 16 cm and 12 cm. Then, the length of the sides of the rhombus is
 - a. 9 cm
 - b. 10 cm
 - c. 8 cm
 - d. 20 cm
7. If the circumference of a circle and the perimeter of a square are equal, then
 - a. Area of the circle = Area of the square
 - b. Area of the circle > Area of the square
 - c. Area of the circle < Area of the square
 - d. Nothing definite can be said about the relation between the areas of the circle & square.
8. In the following figure, two line segments AC and BD intersect each other at the point P such that PA = 6 cm, PB = 3 cm, PC = 2.5 cm, PD = 5 cm, $\angle APB = 50^\circ$ and $\angle CDP = 30^\circ$. Then, $\angle PBA$ is equal to
 - a. 50°
 - b. 30°
 - c. 60°
 - d. 100°
9. If $\cos(\alpha + \beta) = 0$, then $\sin(\alpha - \beta)$ can be reduced to
 - a. $\cos 2\beta$
 - b. $\cos \beta$
 - c. $\sin \alpha$
 - d. $\sin 2\alpha$



10. Which of the following equations has the sum of its roots as 3?

a. $2x^2 - 3x + 6 = 0$

b. $-x^2 + 3x - 3 = 0$

c. $\sqrt{2}x^2 - \frac{3}{\sqrt{2}}x + 1 = 0$

d. $3x^2 - 3x + 3 = 0$

11. What is the next number in the sequence: 1, 2, 5, 12, 27, 58,?

a. 116

b. 121

c. 125

d. 127

12. Gopu ranks sixth from the top and twenty seventh from the bottom in a class. How many students are there in the class?

a. 31

b. 32

c. 33

d. 34

13. Pointing towards a person, a man said to a woman, "his mother is the only daughter of your father."

How is woman related to that person?

a. Daughter

b. Sister

c. Mother

d. Wife

14. p, q, r, s and t are seated around a circular table facing the centre:

i. 't' sits second to the right of 's'.

ii. 'r' sits second to the right of 't'.

iii. 'p' is not an immediate neighbour of 's'.

Who sits between 's' and 't'?

a. q

b. r

c. p

d. t

15. The total age of 'a', 'b' and 'c' is 93 years. Ten years ago the ratio of their ages was 2 : 3 : 4. Then the present age of 'c' is

a. 18

b. 28

c. 38

d. 48

16. $31 : 69 :: 351 : ?$

a. 496

b. 511

c. 521

d. 707

17. If HEALTH is written as GSKZDG, then how will NORTH be written in that code?

a. OPSUI

b. GSQNM

c. FRPML

d. IUSPO

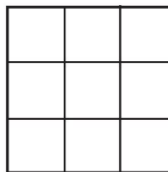
18. How many squares are there in the figure?

a. 18

b. 14

c. 10

d. 9



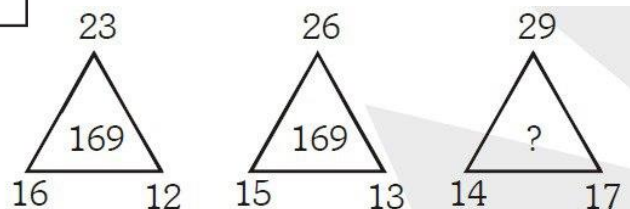
19. What is the missing number?

a. 169

b. 267

c. 209

d. 389



20. Five bells begin to toll together and toll respectively at intervals of 6,5,7,10 and 12 seconds. How many times will they toll together in one hour excluding the one at the beginning?

a. 7

c. 9

b. 8

d. 11

21. How many integers are there between 1 and 100 which have 4 as a digit but are not divisible by 4?

a. 5

b. 11

c. 12

d. 13

22. If 'when' means 'x', 'you' means 'divide', 'come' means '-' and 'will' means '+', then what will be the value of "8 when 12 will 16 you 2 come 10"?

a. 94

b. 45

c. 112

d. 96

23. In a party of 100 people, 60 had Vanilla ice-cream and 40 had chocolate ice-cream. 10 people did not have any ice-cream. How many had only chocolate ice-cream?

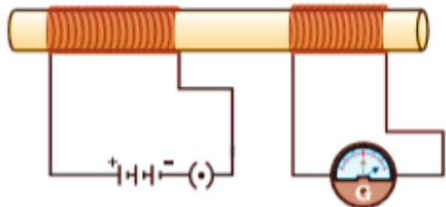
a. 10

b. 0

c. 20

d. 30

24. Satish was born on Aug 15th, Wednesday. If Mahesh is 17 days older than Satish. On which day was Mahesh born?
- a. Tuesday b. Sunday c. Wednesday d. Monday
25. What is the angle between two hands, (hour hand and minute hand) at 10'o clock.?
- a. 30° b. 60° c. 90° d. 120°
26. In the arrangement shown in Figure, there are two coils wound on a non- conducting cylindrical rod. Initially, the key is not inserted. Then the key is inserted and later removed. Then,



- a. The deflection in the galvanometer remains zero throughout
- b. There is a momentary deflection in the galvanometer but it dies out shortly and there is no effect when the key is removed
- c. There are momentary galvanometer deflections that die out shortly; the deflections are in the same direction
- d. There are momentary galvanometer deflections that die out shortly; the deflections are in opposite directions
27. A 10 mm long awl pin is placed vertically in front of a concave mirror. A 5 mm long image of the awl pin is formed at 30 cm in front of the mirror. The focal length of this mirror is:
- a. -20 cm b. -30 cm c. -40 cm d. -60 cm
28. Which of the following statements is correct regarding the propagation of light of different colours of white light in air?
- a. Red light moves fastest
- b. Blue light moves faster than green light
- c. All the colours of the white light move with the same speed
- d. Yellow light moves with the mean speed as that of the red and the violet light
29. A sample of soil is mixed with water and allowed to settle. The clear supernatant solution turns the pH paper yellowish-orange. Which of the following would change the colour of this pH paper to greenish-blue?
- a. Lemon juice c. Common salt
- b. Vinegar d. An antacid
30. Stainless steel is a very useful material for our life. In stainless steel, iron is mixed with:
- a. Ni and Cr b. Cu and Cr c. Ni and Cu d. Cu and Au
31. Find a, b, c, d
- $a \text{ CO}_2 + b \text{ H}_2\text{O} \rightarrow c \text{ C}_6\text{H}_{12}\text{O}_6 + d \text{ O}_2$
- a. a = 3, b = 3, c = 2, d = 4 c. a = 6, b = 6, c = 1, d = 6
- b. a = 4, b = 2, c = 4, d = 2 d. a = 6, b = 4, c = 2, d = 4
32. The maleness of a child is determined by
- a. the X chromosome in the zygote
- b. the Y chromosome in zygote
- c. the cytoplasm of germ cell which determines the sex
- d. sex is determined by chance
33. In a food chain, Grass > Deer > Tiger, Deer is
- a. Secondary Consumer c. Decomposers
- b. Primary Consumer d. Producer

34. Expand the abbreviation GAP
- Governmental Agency for Pollution Control
 - Gross Assimilation by Photosynthesis
 - Ganga Action Plan
 - Governmental Agency for Animal Protection
35. Select the group which shares maximum number of common characters
- two individuals of a species
 - two species of a genus
 - two genera of a family
 - two genera of two families
36. Which India-origin scientist has been awarded UK's royal Order of Merit?
- Venkatraman Ramakrishnan
 - Krishna Chatterjee
 - Subhash Khot
 - Yadvinder Malhi
37. When is the 'National Maritime Day' celebrated in India?
- April 2
 - April 5
 - April 7
 - April 10
38. Below given are some hints on X. Identify X?
- ❖ Born in Saudi Arabia.
 - ❖ Educationist, author and theologian; scholar of Arabic.
 - ❖ Congress leader, active in the national movement.
 - ❖ Education Minister in the first union cabinet
- Maulana Muhammed Ali Jawhar
 - Maulana Abul Kalam Azad
 - Sir Syed Ahmed Khan
 - Muhammed Ali Jinnah
39. The 'Drain Theory' was propounded by
- R. P. Dutt
 - C. R. Das
 - M. N. Roy
 - Dadabhai Naoroji
40. Match the List I with List II and select the correct response from the options given thereafter.
- | List – I | List – II |
|-------------------------------|-------------------------------|
| a) Decameron | 1) Dante |
| b) Divine Comedy | 2) Thomas More |
| c) The Canterbury | 3) Boccaccio Tales |
| d) Utopia | 4) Jeffrey Chaucer |
| a. 2 – a, 3 – b, 1 – c, 4 – d | c. 4 – a, 1 – b, 3 – c, 2 – d |
| b. 3 – a, 1 – b, 4 – c, 2 – d | d. 1 – a, 3 – b, 2 – c, 4 – d |
41. Lascaux cave is situated in
- Spain
 - Turkey
 - England
 - France
42. The Muvendars of South India did not include
- The Cheras
 - The Pandiyas
 - The Chalukyas
 - The Cholas
43. The term 'Hieroglyphics' means
- Wedge shaped
 - Fertile crescent
 - Holy land
 - Holy writings
44. Which city is the host of the 36th National Games of India, 2022?
- Roorkee
 - Bengaluru
 - Ahmedabad
 - Chennai
45. Which institution published the book titled 'India: The Mother of Democracy'?
- Indian Council of Historical Research
 - Indian Council for Cultural Relations
 - Election Commission of India
 - NITI Aayog
46. The instrument used for measuring atmospheric temperature
- Barometer
 - Hygrometer
 - Thermometer
 - Calorimeter

47. Which of the following Indian States has unicameral legislature?
- Bihar
 - Uttar Pradesh
 - Maharashtra
 - Kerala
48. Examine the following statements and identify the correct response from the given options:
 Statement I: Fatehpur Sikri was founded by Akbar.
 Statement II: Akbar introduced the Mansabdari system in the Mughal Empire.
- Statement I is true and statement II is false
 - Statement I is false and statement II is true
 - Both statements are true and statement II is the correct explanation of statement I
 - Both statements are true but statement II is not the correct explanation of statement I
49. The layer in a partially molten state seen just beneath the lithosphere
- Aesthenosphere
 - Inner core
 - Barrysphere
 - Outer core
50. Who was the first Indian women to win Wisden Cricketer of the year?
- Harmanpreet Kaur
 - Mithali Raj
 - Smriti Mandhana
 - Sikha Pandey
51. Against whom did the weavers in Silesia revolt in 1845?
- The policemen
 - Contractors
 - Land owners
 - Politicians
52. Who improved the 'Steam Engine' produced by Newcomen?
- Marco Polo
 - Richard Arkwright
 - James Watt
 - None of these
53. Another name for black soil is _____.
- Khadar
 - Regur
 - Laterite
 - Bangar
54. Which of the following river projects provide hydroelectricity to the industries of Maharashtra?
- Koyna Project
 - Hirakud Project
 - Nagarjuna Sagar Project
 - Ghataprabha Project
55. Which language was declared as the only official language of Sri Lanka by an Act passed in 1956?
- Tamil
 - Sinhala
 - Hindi
 - English
56. What is a signed document submitted to an authorized officer regarding one's personal information called?
- Affidavit
 - Declaration
 - Agreement
 - Appeal
57. The basic services like transportation, banking, insurance, etc., belong to which of the following sectors? (A)
- Primary sector
 - Secondary sector
 - Tertiary sector
 - IT sector
58. Grameen Bank of Bangladesh was founded by _____.
- Prof. M. Yunus
 - Prof. A. Sen
 - Dr. M. M. Singh
 - Ms T. Nasreen
59. Match the columns and select the correct option.
- | Column I | Column II |
|----------------------|---|
| (a) Independence | (i) 1948 of Sri Lanka |
| (b) Belgian | (ii) 1950s and 1960s constitution amendments |
| (c) Conflict between | (iii) 1970 -1993 French and Dutch speaking communities in Belgium |
- (a) - (i), (b) - (ii), (c) - (iii)
 - (a) - (iii), (b) - (i), (c) - (ii)
 - (a) - (ii), (b) - (i), (c) - (iii)
 - (a) - (i), (b) - (iii), (c) - (ii)
60. Select the incorrect match.
- Hind Swaraj – Mahatma Gandhi
 - Chauri Chaura – Halting of NonCooperation Movement
 - Swaraj Party – C.R. Das and Motilal Nehru
 - Rowlatt Act – Before the First World Wa
61. I _____ working all afternoon and have just finished the assignment.
- have been
 - had been
 - shall
 - been

62. Rohan _____ the movie before he read the review.
- a. watches b. have watched c. had watched d. was watching
63. He _____ in the States but he still does not have a command over the English language.
- a. have been living c. have lived
b. has been living d. living
64. By the next month, we shall _____ the project.
- a. has completed c. completed
b. completing d. have completed
65. Every boy and girl _____ in the class today.
- a. are present b. is present c. have present d. had present
66. He _____ daily for a year now.
- a. exercises c. has been exercising
b. was exercising d. have been exercising
67. I _____ this book since morning.
- a. had been reading c. have had read
b. has been reading d. shall be reading
68. Which tense is used to express general truths and facts?
- a. Present continuous tense c. Past perfect tense
b. Present perfect tense d. Present indefinite tense
69. According to the prevailing rate, two dozen _____ rupees one hundred.
- a. costs b. cost c. costing d. costed
e.
70. The Council _____ made its decision.
- a. have b. have had c. has d. having
